

Brand Management Agent

Adobe Express AI Assistant - Internship Project Report

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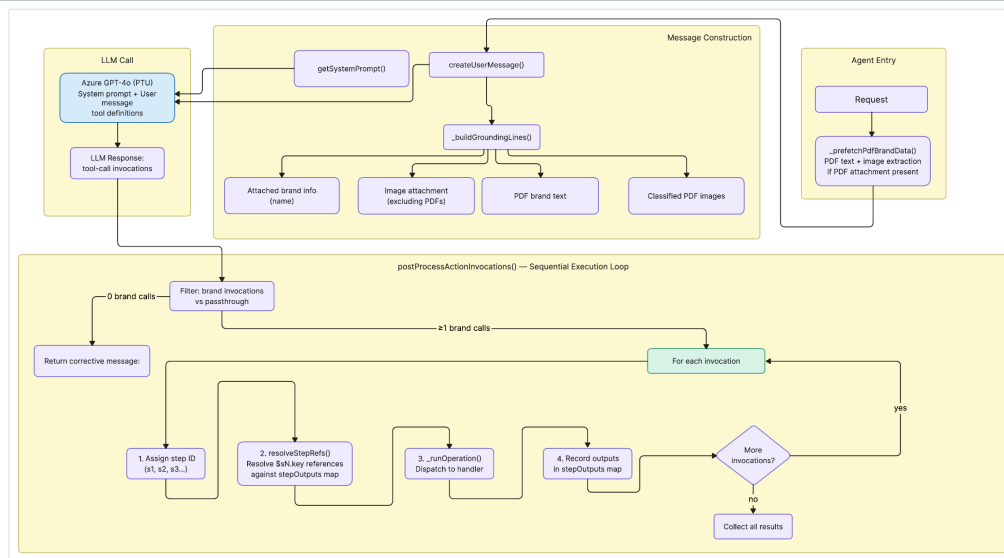
Overview

In Adobe Express, managing a brand still meant leaving the AI Assistant chat for the Brands hub. I built a conversational Brand Management Agent that brings the full brand lifecycle - creating and editing brand libraries, uploading and organizing assets, applying a brand to an open design, and generating on-brand designs - directly into the Assistant. It ships behind the `aiAssistantBrandManagementAgent` feature flag.

Approach

- A standalone Brand Agent prototype was ported into the Horizon monorepo as a conversational subagent alongside `generateTemplatesAgent`, `designQueryAgent`, and `findAndPreviewAssetsAgent`, using the standard Atomic Steps Orchestrator path.
- It reuses Horizon's `MelvilleController` (@hz/brands-core) with the real user token rather than introducing a separate API client; the design is fully agentic, with the LLM driving every action through tool calls and repo conventions throughout.
- Shared routing is flag-gated: `ConversationalIntentAnalysisBlock` classifies brand requests as converse, `PromptCompletenessBlock` treats them as complete, and `ConversationalPlannerBlock` keeps a compound brand request together as one atomic step.

Architecture



Agent surface. 23 LLM tools across three levels: library (create/ rename/ delete/ list, brand guidelines); element (upload/ list/ rename/ delete/ duplicate/ copy-move, primary/ secondary/ header/ body roles, fonts, Sensei color extraction, usage instructions); and design integration.

Multi-step chaining. `stepRefs.ts` lets a later tool call reference an earlier result via `$$N.key` tokens, resolved against a `stepOutputs` map in a sequential execution loop. This supports end-to-end turns such as create brand -> upload logo -> extract colours -> save palette.

PDF brand kit. A brand-guidelines PDF triggers text and image extraction; GPT-4o model classifies images as logo or graphic, then colours, fonts, and usage notes are configured in one pass.

Applying Brand to Designs. `brandApply` uses deterministic recolor and font-swap engines; `brandAddElementToDesign` places an exact full-quality logo/graphic; `brandPlaceLogoOnDesign` uses vision-based placement coordinates.

Feature flags. Gated by four independent flags: `aiAssistantBrandManagementAgent` (main agent gate), `aiAssistantBrandsEnabled` (brands panel visibility), `aiAssistantBrandAssetsInGeneration` (logo reference in generation), and `aiAssistantBrandLogoPlaceholder` (direct logo canvas insertion).

Generation-time context. `GenerateTemplatesSubagent` resolves the target brand (picker-attached or named in free text) through `BrandContextResolver`, carrying logos, graphics, colours, fonts, guidelines, and usage notes.

Generative reference. The primary logo is fetched at 2048 px / full rendition and uploaded as a generative reference image; the brand context is serialized as JSON and appended to the prompt, with the system prompt teaching the model how to use it.

Results. Among `LegoDesigns`, `Firefly3PNb`, and `Firefly3PGptImg2`, `Firefly3PGptImg2` performed best. At 2048 px / full rendition, symbols and shapes reconstruct accurately; wordmark text still deteriorates, reflecting a limitation of generative reference-image reconstruction.

Next step. Embed the original logo directly onto the canvas at generation time to preserve exact wordmark fidelity. And also I propose two new metadata fields to store 'brand guidelines' (stores info about the brand their target audience and design style) and a complete 'asset usage' notes (it provides more info about the placement of that img at the time of generation, etc) to persist more info about a brand and to send those at the time of generation.

Colour extraction. The `extractColorsFromLogo` tool sends a brand logo to Adobe Sensei for dominant-colour analysis. Returned colours are auto-mapped to primary and secondary brand-colour slots, giving users an instant palette from any uploaded logo without manual colour picking.

Image deduplication. SHA-256 hashing deduplicates images extracted from brand-guidelines PDFs before vision classification, preventing redundant uploads and reducing API calls during automated brand-kit setup.